Module 3: Vector space and Linear Transformations:

Introduction :—

Group :- A non empty set equipped with one binary operation $*$ is called group if $*$ satisfies following postulates (or axioms).

1. Closure Property : $\forall$ a and $b \in G \Rightarrow a * b \in G$.
2. Associative Property : $\forall a, b, c \in G \Rightarrow a*(b*c) = (a*b)*c$
3. Existence of identity :
   There exists an element $e \in G$ such that
   $$a*e = a = e*a.$$
   The element e is called the identity.
4. Existence of inverse :
   Each element of G possesses inverse, i.e $\forall a \in G \Rightarrow$ there exists an element $b \in G$ such that
   $$a*b = e = b*a$$
   Then b is called inverse of a and we write $b = a^{-1}$
   $$\therefore a*a^{-1} = e = a^{-1}*a.$$

$*$ A group with commutative property is known as abelian group or commutative group.
i.e $\forall a, b \in G$, if $a*b = b*a$.
$\therefore$ '$*$' is commutative on G.

Eg:- i) $(I, +)$ is an abelian group as $(I,+)$ is a group and also $\forall a, b \in I$.

ii) $(Q_0, \cdot)$, $(R_0, \cdot)$, $(C_0, \cdot)$ are examples of abelian group.

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Rings:-

Suppose R is a non-empty set equipped with two binary operation called addition and multiplication and denoted by '+' and '$\cdot$' respectively.

Then the algebraic structure $(R, +, \cdot)$ is known as ring if the following axioms are satisfied.

1) $(R, +)$ is an abelian group.

   i) closure property for '+'.
   $\forall\ a, b \in R \Rightarrow a+b \in R$.

   ii) Associative property for '+'
   $\forall\ a, b, c \in R \Rightarrow a+(b+c) = (a+b)+c$

   iii) Existence of identity element for '+'
   $\forall\ a \in R\ \exists\ 0 \in R \ni: a+0 = a = 0+a$.
   0 is known as additive element or Zero element of the ring.

   iv) Existence of inverse element for '+'.
   $\forall\ a \in R\ \exists\ -a \in R$.
   such that $a+(-a) = 0 = -a+a$.
   $-a$ is known as additive inverse of a.

2) $(R, \cdot)$ is a semi group.

   i) closure property for '$\cdot$'
   $\forall\ a, b \in R \Rightarrow a \cdot b \in R$.

   ii) Associative property for '$\cdot$'
   $\forall\ a, b, c \in R \Rightarrow a \cdot (b \cdot c) = (a \cdot b) \cdot c$

3) Multiplication '$\cdot$' distributes over addition '+' from left and also from right.
   $\forall\ a, b, c \in R$
   i) $a \cdot (b+c) = a \cdot b + a \cdot c$
   ii) $(a+b) \cdot c = a \cdot c + b \cdot c$

Eg:- $(I, +, \cdot)$ is a ring.

* Commutative Ring:-

Ring $(R, +, \cdot)$ is a commutative ring, if $\forall a, b \in R$, $a \cdot b = b \cdot a$.

Eg:- $(I, +, \cdot)$.

* Ring With Unity :-

Ring $(R, +, \cdot)$ is known as ring with unity if $\forall a \in R \; \exists 1 \in R$ such that $a \cdot 1 = a = 1 \cdot a$.

* Commutative Ring with Unity:-

Ring $(R, +, \cdot)$ is commutative ring with unity, if $\forall a, b \in R \Rightarrow a \cdot b = b \cdot a$ and $\forall a \in R \; \exists 1 \in R$ such that $a \cdot 1 = a = 1 \cdot a$.

Field :-

A commutative ring with unity in which every non-zero element possesses their multiplicative inverse is known as field.

i.e $(F, +, \cdot)$ is a field if

(i) $(F, +)$ is an abelian group.

(ii) $(F_0, \cdot)$ is an abelian group.

(iii) Multiplication distributes over addition from the left and also from the right.

Eg:- $(Q, +, \cdot)$, $(R, +, \cdot)$, $(C, +, \cdot)$ are examples of fields.

*** Note :-**

If $(F, +, \cdot)$ is a field then F contains at least two elements, Zero element and unit element i.e additive identity element and multiplicative identity element.

Vector Spaces:-

Let F be a field. A vector space over F, is a non-empty set V together with two operations (called addition and scalar multiplication) such that for each $u, v \in V$ there is a unique element $u+v \in V$ and for each $\alpha \in F$ and $u \in V$ there is a unique element $\alpha u \in V$, and it satisfies the following conditions:

I. (i) $u+(v+w) = (u+v)+w$, for all $u, v, w \in V$ (Associativity)

(ii) $u+v = v+u$, for all $u, v \in V$ (Commutativity)

(iii) There exists an element $0 \in V$ such that $u+0 = u = 0+u$, for all $u \in V$ (Existence of identity element)

(iv) For each $u \in V$, there exists a unique element $-u \in V$ such that $u+(-u) = 0 = (-u)+u$ (Existence of additive inverse)

II. There is an external composition in V over F called scalar multiplication.
i.e $\forall \alpha \in F$ and $u \in V \Rightarrow \alpha . u \notin V$
In otherwords V is closed with respect to scalar multiplication.

III. The two compositions, i.e. vector addition and scalar multiplication satisfy the following postulates,

$\forall\ a, b \in F$ and $v, w \in V$

(i) $(a+b) \cdot v = a \cdot v + b \cdot v$

(ii) $a \cdot (b \cdot v) = (ab) \cdot v$

(iii) $a \cdot (v+w) = a \cdot v + a \cdot w$

(iv) $1 \cdot (v) = v$.

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* Vectors in R^n :-

The set of all ordered triples (a, b, c) of real numbers is called Euclidean 3-space and is denoted by R^3 & the set of all n-tuples of real numbers, denoted by R^n, is called Euclidean n-space.

Eg:- The ordered pair $(2, -3)$ belongs to R^2; it is a 2-tuple of dimension two.

The ordered triple $(7, 3, 6)$ belongs to R^3; it is a 3-tuple of dimension three.

Examples or Problems :-

1) Prove that the set of all vectors in a plane over the field of real numbers is a vector space with respect to vector addition and scalar multiplication.

Solⁿ:- Let V denote the set of all coplanar vectors and R be the field of real numbers.

$\therefore$ The elements of V are the ordered pairs (x, y) where $x, y \in R$.

$$V = \{(x, y) : x, y \in R\}$$

I) $(V, +)$ is an abelian group.

i) Associativity :-

We know that for all $u, v, w \in V$

$$(u+v)+w = u+(v+w).$$

ii) Commutativity :-

We know that for all $u, v \in V$

$$u+v = v+u$$

iii) Existence of additive identity :-

For every vector $u \in V$ there exists a zero vector $O \in V$ such that $u+O = O+u = u$.

iv) Existence of additive inverse :-

For every vector $u \in V$ there exists a vector $-u \in V$ such that $u+(-u) = (-u)+u = O$.

Thus V is an abelian group with respect to vector addition.

II) Scalar multiplication in V.

i) For $u, v \in V$ and $\alpha \in R$ we have

$$\alpha(u+v) = \alpha u + \alpha v$$

ii) For $u \in V$ and $a, b \in R$, we have

$$(a+b)u = au + bu.$$

iii) For $u \in V$ and $a, b \in R$, we have

$$a(bu) = (ab)u$$

iv) For $u \in V$ and $1 \in R$, we have

$$1u = u.$$

Thus set V of coplanar vectors satisfies all the properties of vector addition and scalar multiplication.

$\therefore$ V is a vector space.

Q) Prove that the set C of all complex numbers (i.e the set of all ordered pairs of real numbers) is a vector space over the field R of all real numbers where vector addition is defined by

$$(x_1, x_2) + (y_1, y_2) = (x_1 + y_1, x_2 + y_2), \text{ for all } (x_1, x_2), (y_1, y_2) \in C$$

and scalar multiplication is defined by

$$\alpha(x_1, x_2) = (\alpha x_1, \alpha x_2), \text{ for all } \alpha \in R.$$

Soln:- We observe that C is closed under vector addition and under scalar multiplication.

I. $(C, +)$ is an abelian group.

(i) Associativity:

For all $(x_1, x_2), (y_1, y_2), (z_1, z_2) \in C$, we have

$$\begin{aligned}(x_1, x_2) + [(y_1, y_2) + (z_1, z_2)] &= (x_1, x_2) + (y_1 + z_1, y_2 + z_2) \\ &= (x_1 + y_1 + z_1, x_2 + y_2 + z_2) \\ &= (x_1 + y_1, x_2 + y_2) + (z_1, z_2) \\ &= [(x_1, x_2) + (y_1, y_2)] + (z_1, z_2)\end{aligned}$$

(ii) Commutativity:

For all $(x_1, x_2), (y_1, y_2) \in C$, we have

$$\begin{aligned}(x_1, x_2) + (y_1, y_2) &= (x_1 + y_1, x_2 + y_2) \\ &= (y_1 + x_1, y_2 + x_2) \\ &= (y_1, y_2) + (x_1, x_2)\end{aligned}$$

(iii) Existence of identity:

For $(x_1, x_2) \in C$, there exists $(0,0) \in C$ such that

$$\begin{aligned}(x_1 + x_2) + (0,0) &= (x_1 + 0, x_2 + 0) \\ &= (x_1, x_2)\end{aligned}$$

$$(0+0) + (x_1, x_2) = (0 + x_1, 0 + x_2) = (x_1, x_2)$$

$$\therefore (x_1, x_2) + (0,0) = (x_1, x_2) = (0,0) + (x_1, x_2),$$

(iv) Existence of inverse :

For any $(x_1, x_2) \in G$, there exists $(-x_1, -x_2) \in G$ such that

$$(x_1, x_2) + (-x_1, -x_2) = (0, 0) = (-x_1, -x_2) + (x_1, x_2).$$

Thus C is an abelian group with respect to vector addition.

II. Properties of Scalar multiplication in C.

(i)
$$\begin{aligned}\alpha[(x_1, x_2) + (y_1, y_2)] &= \alpha(x_1 + y_1, x_2 + y_2)\\ &= (\alpha(x_1 + y_1), \alpha(x_2 + y_2))\\ &= (\alpha x_1 + \alpha y_1, \alpha x_2 + \alpha y_2)\\ &= (\alpha x_1, \alpha x_2) + (\alpha y_1, \alpha y_2)\\ &= \alpha(x_1, x_2) + \alpha(y_1, y_2)\end{aligned}$$

Thus $\alpha[(x_1, x_2) + (y_1, y_2)] = \alpha(x_1, x_2) + \alpha(y_1, y_2)$ for all $(x_1, x_2), (y_1 y_2) \in G, \alpha \in R$.

(ii)
$$\begin{aligned}(a+b)(x_1, x_2) &= ((a+b)x_1, (a+b)x_2)\\ &= (ax_1 + bx_1, ax_2 + bx_2)\\ &= (ax_1, ax_2) + (bx_1, bx_2)\\ &= a(x_1, x_2) + b(x_1, x_2)\end{aligned}$$

Thus $(a+b)(x_1, x_2) = a(x_1, x_2) + b(x_1, x_2)$ for all $(x_1, x_2), (y_1, y_2) \in G$ & $a, b \in R$.

(iii) $a(b(x_1, x_2)) = a(bx_1, bx_2) = (abx_1, abx_2)$

$a(b(x_1, x_2)) = (ab)(x_1, x_2)$

(iv) $1.(x_1, x_2) = (x_1, x_2)$ for all $(x_1, x_2) \in G$

Thus the set G satisfies all the properties of vector space.

Hence G is a vector space over R.

3) Show that the set V of all real valued continuous functions of x defined on interval $[0,1]$ is a vector space over the field R of real numbers with respect to vector addition and scalar multiplication defined by

$$(f_1+f_2)x = f_1(x)+f_2(x), \text{ for all } f_1, f_2 \in V$$

$$(\alpha f_1)x = \alpha f_1(x), \text{ for all } \alpha \in R, f_1 \in V.$$

Soln:-

I. $(V,+)$ is an abelian group.

(i) Associativity:

Let $f_1, f_2, f_3 \in V$ be arbitrary.

$$\begin{aligned}[(f_1+f_2)+f_3](x) &= (f_1+f_2)(x)+f_3(x)\\ &= [f_1(x)+f_2(x)]+f_3(x)\\ &= f_1(x)+[f_2(x)+f_3(x)]\\ &= f_1(x)+(f_2+f_3)(x)\\ &= [f_1+(f_2+f_3)](x)\end{aligned}$$

$$(f_1+f_2)+f_3 = f_1+(f_2+f_3).$$

(ii) Commutativity:

Let $f_1, f_2 \in V$ be arbitrary.

$$\begin{aligned}(f_1+f_2)(x) &= f_1(x)+f_2(x)\\ &= f_2(x)+f_1(x)\\ &= (f_2+f_1)(x), \text{ for all } x.\end{aligned}$$

$$f_1+f_2 = f_2+f_1.$$

(iii) Existence of Identity:

Define a function 0 such that $0(x)=0$ (real number), for all $x \in [0,1]$.

Also, 0 is a continuous function and belongs to V

$$(0+f_1)(x) = 0(x)+f_1(x) = 0+f_1(x) = f_1(x)$$

$$(f_1+0)(x) = f_1(x)+0(x) = f_1(x)$$

$$0+f_1 = f_1+0 = f_1$$

Thus, the function O defined above is an additive identity element of V.

(iv) <u>Existence of Inverse</u>:

For a function f_1, the function $-f_1$ defined by $(-f_1)(x) = -f_1(x)$, is called additive inverse, as $[f_1+(-f_1)](x) = f_1(x) + (-f_1)(x)$

$$= f_1(x) - f_1(x) = 0 = O(x)$$

111^{ly}, $[-f_1+f_1](x) = O(x)$.

Thus the set V is an abelian group under addition.

II. <u>Properties of Scalar Multiplication in V.</u>

(i) For $f_1, f_2 \in V$ and $\alpha \in R$, We have

$$[\alpha(f_1+f_2)]x = \alpha[(f_1+f_2)(x)]$$
$$= \alpha[f_1(x)+f_2(x)]$$
$$= \alpha f_1(x) + \alpha f_2(x)$$
$$= (\alpha f_1)x + (\alpha f_2)x$$
$$= (\alpha f_1 + \alpha f_2)x$$
$$\alpha(f_1+f_2) = \alpha f_1 + \alpha f_2$$

(ii) For $f_1 \in V$ and $a, b \in R$.

$$[(a+b)f_1](x) = (a+b)f_1(x)$$
$$= a f_1(x) + b f_1(x)$$
$$= (a f_1)x + (b f_1)x$$
$$= (a f_1 + b f_1)(x)$$
$$(a+b)f_1 = a f_1 + b f_1$$

(iii) For $f_1 \in V$ and $a, b \in R$,

$$[a(bf_1)]x = a[(bf_1)x]$$
$$= a[b f_1(x)]$$
$$= (ab)f_1(x)$$
$$= [(ab)f_1]x$$
$$a(bf_1) = (ab)f_1$$

(iv) For $f_1 \in V$ and $1 \in R$, We have

$$(1.f_1)x = 1.f_1(x) = f_1(x)$$

$$1 f_1 = f_1 .$$

$\therefore$ Thus V satisfies all the properties of a vector space and hence V is a vector space.

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4) Show that the set of all matrices of the type $m \times n$ where m and n are fixed positive integers is a vector space over R with respect to matrix addition and multiplication of matrix by a scalar (i.e. a real number)

Soln: Let M denotes the set of all matrices of type $m \times n$.

I. $(M,+)$ is an abelian group.

(i) Associativity:

Let $A = [a_{ij}]_{m\times n}$, $B = [b_{ij}]_{m\times n}$, $C = [c_{ij}]_{m\times n}$ be three matrices belonging to set M.

$$\begin{aligned}(A+B)+C &= ([a_{ij}]+[b_{ij}]) + [c_{ij}] \\ &= [a_{ij}+b_{ij}] + [c_{ij}] \\ &= [(a_{ij}+b_{ij}) + (c_{ij})] \\ &= [a_{ij} + (b_{ij}+c_{ij})] \\ &= [a_{ij}] + [b_{ij}+c_{ij}] \\ &= [a_{ij}] + ([b_{ij}]+[c_{ij}]) \\ &= A+(B+C).\end{aligned}$$

(ii) Commutativity:

Let $A = [a_{ij}]_{m\times n}$ and $B = [b_{ij}]_{m\times n}$ be two matrices belonging to set M.

$$A+B=[a_{ij}]+[b_{ij}]=[a_{ij}+b_{ij}]$$
$$=[b_{ij}+a_{ij}]$$
$$=[b_{ij}]_{m\times n}+[a_{ij}]_{m\times n}$$
$$=B+A.$$

(iii) Existence of Identity:

Let $A=[a_{ij}]_{m\times n}\in M$ be arbitrary.

We know that a zero matrix (null matrix) of type $m\times n$ also belongs to the set M and is denoted by O.

Now, $A+O=[a_{ij}]+[0]=[a_{ij}+0]=[a_{ij}]=A$

III^{ly}, $O+A=A$.

(iv) Existence of Inverse:

If $A=[a_{ij}]_{m\times n}$ belongs to the set M, then

$-A=[-a_{ij}]_{m\times n}$ also belongs to the set M.

Now, $A+(-A)=[a_{ij}]+[-a_{ij}]=[a_{ij}-a_{ij}]$
$$=[0]=O$$

III^{ly} $(-A)+(A)=O$

Thus the set M is an abelian group under addition.

II. Properties of Scalar multiplication in M.

(i) Let $A=[a_{ij}]_{m\times n}$ and $B=[b_{ij}]_{m\times n}\in M$ and $\alpha\in R$, then
$$\alpha(A+B)=\alpha([a_{ij}]+[b_{ij}])$$
$$=\alpha[a_{ij}+b_{ij}]$$
$$=[\alpha(a_{ij}+b_{ij})]$$
$$=[\alpha.a_{ij}+\alpha.b_{ij}]$$ [$\because$ Multiplication is distributive in R]
$$=[\alpha.a_{ij}]+[\alpha.b_{ij}]$$
$$=\alpha[a_{ij}]+\alpha[b_{ij}]=\alpha A+\alpha B.$$

(ii) Let $A = [a_{ij}]_{m\times n} \in M$ and $a, b \in R$, then

$$(a+b)A = (a+b)[a_{ij}]$$
$$= [(a+b)a_{ij}]$$
$$= [a.a_{ij} + b.a_{ij}]$$
$$= a[a_{ij}] + b[a_{ij}]$$
$$= aA + bA.$$

(iii) Let $A = [a_{ij}]_{m\times n} \in M$ and $a, b \in R$ then

$$a(bA) = a(b[a_{ij}])$$
$$= a([ba_{ij}])$$
$$= [(ab)a_{ij}]$$
$$= (ab)[a_{ij}]$$
$$= (ab)A$$
$$a(bA) = (ab)A.$$

(iv) Let $A = [a_{ij}]_{m\times n} \in M$, then

$$1.A = 1.[a_{ij}] = [1.a_{ij}] = [a_{ij}] = A.$$

Thus M satisfies all the properties of vector space and hence M is a vector space over R.

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5) If V is an abelian group of positive real number for multiplication. Define scalar multiplication in V by $a \cdot x = x^a$ where $a \in R$, $x \in V$. Show that V is a vector space over R.

Soln:- Since $V = R^+$ is an abelian group ~~of~~ for multiplication.

There are two operations

(i) $\times$, i.e multiplication in R^+

(ii) $\bullet$, i.e Scalar multiplication in R^+

$a \in R,\ x \in R^+$

$a \cdot x = x^a$

and $x^a \in R^+$ as $x \in R^+$, x^a is always positive.

Eg:- Let $a = -3$

$-3 \cdot x = x^{-3} = \frac{1}{x^3}$ is a positive real number.

Hence x^{-3} is an element ~~of~~ R^+.

Now,

I. $(V, \times)$ is an abelian group.

II. $\forall a \in R, x \in V \Rightarrow a \cdot x = x^a \in V$.

III. (i) $(a+b) \cdot x = x^{a+b} = x^a \times x^b = (a \cdot x) \times (b \cdot x)$

(ii) $a \cdot (x_1 \times x_2) = (x_1 \times x_2)^a$

$= x_1^a \times x_2^a$

$= (a \cdot x_1) \times (a \cdot x_2)$

(iii) $a \cdot (b \cdot x) = (b \cdot x)^a$

$= (x^b)^a = x^{ba}$

$= x^{ab}$ ($\because R$ is commutative for multiplication)

$= ab \cdot (x)$.

(iv) $1 \cdot (x) = x^1 = x$.

Hence $(V, \times, \cdot)$ is a vector space.

Subspaces :-

A non-empty subset W of a vector space $V(F)$ is said to form a subspace of V if W is also a vector space over F with the same addition and scalar multiplication as for V.

Eg:- Let $W_1 = \{(a,0,0) : a \in R\}$

$W_2 = \{(a,b,0) : a,b \in R\}$.

Here W_1 is a subspace of W_2. Also W_1 and W_2 are subspace of R^3.

* Necessary and Sufficient conditions for a subspace:

(Only statement, no proof).

Theorem 1: W is a subspace of $V(F)$ iff

i) W is non empty.

ii) W is closed under vector addition.

i.e $\forall w_1, w_2 \in W \Rightarrow w_1 + w_2 \in W$.

iii) W is closed under scalar multiplication.

i.e $\forall a \in F$ and $w \in W \Rightarrow a.w \in W$.

Theorem 2: W is a subspace of $V(F)$ iff

i) W is non empty.

ii) $\forall a, b \in F$ and $v, w \in W \Rightarrow$

$a.v + b.w \in W$.

Problems:-

1) Show that W is a subspace of $V(R)$ where $W=\{f: f(9)=0\}$

Sol^n:- Since $0 \in W$ as $0(9)=0$

So W is non empty set.

Let $f, g \in W$.

i.e $f(9)=0$ and $g(9)=0$

then $\forall a, b \in R$

$$(a.f+b.g)9 = a.f(9)+b.g(9) = a.0+b.0 = 0$$

Hence $a.f+b.g \in W$.

By theorem (2), W is a subspace of $V(R)$.

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2) Show that W is a subspace of $V(R)$ where $W=\{f: f(2)=f(1)\}$

Sol^n:- $0 \in W$ since $0(2)=0=0(1)$

hence W is a non empty set.

Let $f, g \in W$ then $f(2)=f(1)$ & $g(2)=g(1)$

then $\forall a, b \in R$.

$$(a.f+b.g)(2) = a.f(2)+b.g(2) = a.f(1)+b.g(1)$$
$$= (a.f+b.g)(1)$$

hence $a.f+b.g \in W$.

So by theorem 2, W is a subspace of $V(R)$.

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3) Let $V=R^3$ be the Euclidean 3-space. Let $W=\{(x,y,z): ax+by+cz=0;\ x,y,z \in R\}$. a,b,c being real numbers. Show that W is a subspace of V.

(OR)

Show that any plane passing through the origin is a subspace of R^3.

Soln:- Let $u=(x_1,y_1,z_1)$, $v=(x_2,y_2,z_2)$ be any two elements of W where $x_1,x_2,y_1,y_2,z_1,z_2 \in R$.

Then $ax_1+by_1+cz_1=0$ and $ax_2+bx_2+cx_2=0$.

For $\alpha \in R$, $\alpha u+v = \alpha(x_1,y_1,z_1)+(x_2,y_2,z_2)$

$\alpha u+v = (\alpha x_1, \alpha y_1, \alpha z_1)+(x_2,y_2,z_2)$

$= (\alpha x_1+x_2, \alpha y_1+y_2, \alpha z_1+z_2) \rightarrow (1)$

Where αx_1+x_2, αy_1+y_2, $\alpha z_1+z_2 \in R$.

Now

$a(\alpha x_1+x_2)+b(\alpha y_1+y_2)+c(\alpha z_1+z_2) =$

$\alpha(ax_1+by_1+cz_1)+(ax_2+by_2+cz_2)$

$= \alpha(0)+0 = 0. \rightarrow (2)$.

From (1) and (2), we have

$\alpha u+v \in W$.

Thus, W is a subspace of V.

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4) Let V be the vector space of all square matrices over R. Determine which of the following are subspaces of V.

(i) $W=\left\{\begin{bmatrix} x & y \\ z & 0 \end{bmatrix} : x,y,z \in R\right\}$ (ii) $W=\left\{\begin{bmatrix} x & 0 \\ 0 & y \end{bmatrix} : x,y \in R\right\}$

(iii) $W=\{A : A \in V$ and A is singular$\}$ (iv) $W=\{A : A \in V, A^2=A\}$

Solⁿ:- (i) Let $A = \begin{bmatrix} x_1 & y_1 \\ z_1 & 0 \end{bmatrix}$, $B = \begin{bmatrix} x_2 & y_2 \\ z_2 & 0 \end{bmatrix}$ be any two elements of W.

If $a, b \in R$ then

$$aA + bB = a\begin{bmatrix} x_1 & y_1 \\ z_1 & 0 \end{bmatrix} + b\begin{bmatrix} x_2 & y_2 \\ z_2 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} ax_1 & ay_1 \\ az_1 & 0 \end{bmatrix} + \begin{bmatrix} bx_2 & by_2 \\ bz_2 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} ax_1 + bx_2 & ay_1 + by_2 \\ az_1 + bz_2 & 0 \end{bmatrix}$$

which is a matrix of the type $\begin{bmatrix} x & y \\ z & 0 \end{bmatrix}$ and

$ax_1 + bx_2,\ ay_1 + by_2,\ az_1 + bz_2 \in R$.

$\therefore\ aA + bB \in W$.

Thus, W is a sub-space of V.

(ii) Let $A = \begin{bmatrix} x_1 & 0 \\ 0 & y_1 \end{bmatrix}$, $B = \begin{bmatrix} x_2 & 0 \\ 0 & y_2 \end{bmatrix}$ be any two elements of W.

If $a, b \in R$, then

$$aA + bB = a\begin{bmatrix} x_1 & 0 \\ 0 & y_1 \end{bmatrix} + b\begin{bmatrix} x_2 & 0 \\ 0 & y_2 \end{bmatrix}$$

$$= \begin{bmatrix} ax_1 + bx_2 & 0 \\ 0 & ax_2 + by_2 \end{bmatrix}$$

which is a matrix of the type $\begin{bmatrix} x & 0 \\ 0 & y \end{bmatrix}$ and

$ax_1 + bx_2,\ ay_1 + by_2 \in R$.

$\therefore\ aA + bB \in W$.

Thus W is a subspace of V.

(iii) Here W is the set of singular matrices.

Let $A = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$ $B = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$ be two square matrices.

Since $|A| = 0$ and $|B| = 0$. therefore $A, B \in W$.

If $a, b \in R$ are non zero, then

$$aA + bB = a\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} + b\begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} a & 0 \\ 0 & 0 \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ 0 & b \end{bmatrix}$$

$$= \begin{bmatrix} a & 0 \\ 0 & b \end{bmatrix}$$

Also, $|aA_1 + bA_2| = \begin{vmatrix} a & 0 \\ 0 & b \end{vmatrix} = ab \neq 0$, as none of a and b is zero.

$\therefore$ $aA_1 + bA_2 \notin W$.

Thus, W is not a sub-space of V.

(iv) Let $A = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$ so that $A^2 = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} = A$

$\therefore$ $A \in W$.

Now $A + A = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} + \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 2 & 0 \\ 0 & 0 \end{bmatrix}$

$$(A+A)^2 = \begin{bmatrix} 4 & 0 \\ 0 & 0 \end{bmatrix} + \begin{bmatrix} 2 & 0 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 4 & 0 \\ 0 & 0 \end{bmatrix}$$

$$\neq A + A.$$

$\therefore$ $(A + A)$ is not a member of the set W.

$\Rightarrow$ W is not closed under addition.

Hence W is not a sub-space of V.

5) Let V be the vector space of all real valued continuous functions over $\mathbb{R}$. Show that the set W of solutions of differential equations $5\frac{d^2y}{dx^2} - 7\frac{dy}{dx} + 2y = 0$ is a subspace of V.

Sol^n:- Let $y_1, y_2 \in W$. Then y_1, y_2 are solutions of differential equations.

$$5\frac{d^2y}{dx^2} - 7\frac{dy}{dx} + 2y = 0 \quad \rightarrow (1)$$

i.e $\quad 5\frac{d^2y_1}{dx^2} - 7\frac{d^2y_1}{dx} + 2y_1 = 0 \; ; \; 5\frac{d^2y_2}{dx^2} - 7\frac{d^2y_2}{dx^2} + 2y_2 = 0.$

Let $a, b \in \mathbb{R}$ be arbitrary.

$$5\frac{d^2}{dx^2}(ay_1 + by_2) - 7\frac{d}{dx}(ay_1 + by_2) + 2(ay_1 + by_2)$$

$$= 5\frac{d^2}{dx^2}(ay_1) - 7\frac{d}{dx}(ay_1) + 2(ay_1) +$$

$$5\frac{d^2}{dx^2}(by_2) - 7\frac{d}{dx}(by_2) + 2(by_2).$$

$$= a\left[5\frac{d^2y_1}{dx^2} - 7\frac{dy_1}{dx} + 2y_1\right] + b\left[5\frac{d^2y_2}{dx^2} - 7\frac{dy_2}{dx} + 2y_2\right]$$

$$= a.0 + b.0 = 0.$$

Thus $ay_1 + by_2$ is a solution of (1) & so $ay_1 + by_2 \in W$

$\therefore$ W is a subspace of V.

Linear Combination :–

Let V be a vector space over a field F and let $v_1, v_2 \dots v_n \in V$.

Any vector of the form $a_1 \cdot v_1 + a_2 \cdot v_2 + \dots + a_n \cdot v_n$ in V, where $a_i \in F$ is called a linear combination of $v_1, v_2 \dots v_n$.

Linear Dependence :–

Let V be a vector space over the field F. The vectors $v_1, v_2 \dots v_n$ are said to be linearly dependent over F, if $\exists$ scalars $a_1, a_2 \dots a_n \in F$ not all zero but linear combination is zero.

i.e $a_1 \cdot v_1 + a_2 \cdot v_2 + \dots + a_n \cdot v_n = 0$.

but all $a_i \neq 0$, where $i \in N$.

Linear Independence :–

Let V be a vector space over the field F. The vectors $v_1, v_2 \dots v_n$ are said to be linearly independent over F, if $\exists$ scalars $a_1, a_2 \dots a_n \in F$ such that

$a_1 \cdot v_1 + a_2 \cdot v_2 + \dots + a_n \cdot v_n = 0 \Rightarrow$ all $a_i = 0$ where $i \in N$.

Linear Span :–

Let S be a subset of the vector space V over the field F. The set of all linear combinations of vectors in S is called a linear span of S and is denoted by $L(S)$.

If $S = \phi$, then $L(S) = 0$.

Basis or Base of Vector space V :-

Let V be a vector space over the field F. The set of vectors $\{v_1, v_2 ---- v_n\}$ is called a basis of V, if

(i) $v_1, v_2 ---- v_n$ are linearly independent.

(ii) $v_1, v_2 --- v_n$ span V. i.e. each vector of V can be uniquely expressed as linear combination of $v_1, v_2 ---, v_n$.

Dimension of a vector space V :-

Number of elements in a basis of vector space V is called the dimension of V and is denoted by dim V.

If V contains a basis with n elements then the dim V = n.

Note :-

1) The vector space $\{0\}$ is defined to have dim 0, since Empty set ϕ is independent and generates $\{0\}$.

$\therefore$ dim $\{0\}$ = Number of elements in ϕ

= 0 [Since no element is in ϕ]

2) When a vector space is not of finite dimension, it is said to be of infinite dimension.

Problems :—

1) Express the vector $v=(1,-2,5)$ as a linear combination of the vectors $v_1=(1,1,1)$, $v_2=(1,2,3)$, $v_3=(2,-1,1)$ in the vector space $R^3(R)$.

Solⁿ :- Let $v = a_1 v_1 + a_2 v_2 + a_3 v_3$; $a_1, a_2, a_3 \in R$

$$(1,-2,5) = a_1(1,1,1) + a_2(1,2,3) + a_3(2,-1,1)$$

$$(1,-2,5) = (a_1+a_2+2a_3,\ a_1+2a_2-a_3,\ a_1+3a_2+a_3)$$

Equating the corresponding elements, we get

$$a_1+a_2+2a_3=1\ ;\ a_1+2a_2-a_3=-2\ ;\ a_1+3a_2+a_3=5$$

Solving above equations, we get

$$a_1=-6,\quad a_2=3,\quad a_3=2$$

Hence $(1,-2,5) = -6(1,1,1)+3(1,2,3)+2(2,-1,1)$.

---

2) Write the vector $v=(1,3,9)$ as a linear combination of the vectors $u_1=(2,1,3)$, $u_2=(1,-1,1)$, $u_3=(3,1,5)$

(Another method of solving)

Solⁿ :- Let $v = xu_1 + yu_2 + zu_3$

$$(1,3,9) = x(2,1,3) + y(1,-1,1) + z(3,1,5)$$

$$\Rightarrow\quad 2x+y+3z=1\ ;\ x-y+z=3\ ;\ 3x+y+5z=9$$

Consider $AX=B$.

$$A=\begin{bmatrix}2 & 1 & 3\\ 1 & -1 & 1\\ 3 & 1 & 5\end{bmatrix}\quad X=\begin{bmatrix}x\\ y\\ z\end{bmatrix}\quad B=\begin{bmatrix}1\\ 3\\ 9\end{bmatrix}$$

Augmented matrix,

$$[A:B] = \begin{bmatrix} 2 & 1 & 3 & : & 1 \\ 1 & -1 & 1 & : & 3 \\ 3 & 1 & 5 & : & 9 \end{bmatrix}$$

$R_1 \leftrightarrow R_2$

$$[A:B] = \begin{bmatrix} 1 & -1 & 1 & : & 3 \\ 2 & 1 & 3 & : & 1 \\ 3 & 1 & 5 & : & 9 \end{bmatrix}$$

$R_2 \to R_2 - 2R_1$

$R_3 \to R_3 - 3R_1$

$$[A:B] = \begin{bmatrix} 1 & -1 & 1 & : & 3 \\ 0 & 3 & 1 & : & -5 \\ 0 & 4 & 2 & : & 0 \end{bmatrix}$$

$R_3 \to R_3 - R_2$

$$[A:B] = \begin{bmatrix} 1 & -1 & 1 & : & 3 \\ 0 & 3 & 1 & : & -5 \\ 0 & 1 & 1 & : & 5 \end{bmatrix}$$

$R_3 \leftrightarrow R_4$

$$[A:B] = \begin{bmatrix} 1 & -1 & 1 & : & 3 \\ 0 & 1 & 1 & : & +5 \\ 0 & 3 & 1 & : & -5 \end{bmatrix}$$

$R_3 \to R_3 - 3R_2$

$$[A:B] = \begin{bmatrix} 1 & -1 & 1 & : & 3 \\ 0 & 1 & 1 & : & 5 \\ 0 & 0 & -2 & : & -20 \end{bmatrix}$$

$\therefore$ Rank $[A]$ = Rank $[A:B]$ = 3 = Number of unknowns

$\therefore$ The system of linear equations is consistent and Possesses a unique soln.

$x+y+z=3$; $y+z=5$; $-2z=-20$

$\therefore$ $z=10$, $y=-5$, $x=-12$

$\therefore \quad (1,3,9) = -12(2,1,3) - 5(1,-1,1) + 10(3,1,5).$

---

3) Write the vector $v = (4,2,1)$ as a linear combination of the vectors $u_1 = (1,-3,1)$, $u_2 = (0,1,2)$, $u_3 = (5,1,37)$.

Sol^n:- Let $v = xu_1 + yu_2 + zu_3$

$$(4,2,1) = x(1,-3,1) + y(0,1,2) + z(5,1,37)$$

$$\Rightarrow \begin{aligned} x + 0y + 5z &= 4 \\ -3x + y + z &= 2 \\ x + 2y + 37z &= 1 \end{aligned}$$

Let $AX = B$. where $A = \begin{bmatrix} 1 & 0 & 5 \\ -3 & 1 & 1 \\ 1 & 2 & 37 \end{bmatrix}$ $X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$ $B = \begin{bmatrix} 4 \\ 2 \\ 1 \end{bmatrix}$

Consider, $[A:B] = \left[\begin{array}{ccc:c} 1 & 0 & 5 & 4 \\ -3 & 1 & 1 & 2 \\ 1 & 2 & 37 & 1 \end{array}\right]$

$R_2 \to R_2 + 3R_1$
$R_3 \to R_3 - R_1$

$[A:B] = \left[\begin{array}{ccc:c} 1 & 0 & 5 & 4 \\ 0 & 1 & 16 & 14 \\ 0 & 2 & 32 & -3 \end{array}\right]$

$R_3 \to R_3 - 2R_2$

$[A:B] = \left[\begin{array}{ccc:c} 1 & 0 & 5 & 4 \\ 0 & 1 & 16 & 14 \\ 0 & 0 & 0 & -31 \end{array}\right]$

Rank $[A] = 2 \neq$ Rank $[A:B] = 3$.

Since Rank $[A] \neq$ Rank $[A:B]$. Hence the system of linear equations is inconsistent.

i.e sol^n does not exist.

$\therefore$ v cannot be expressed as linear combination of the vectors u_1, u_2, u_3.

4) For what value of k (if any) the vector $v=(1,-2,k)$ Can be expressed as a linear combination of vectors $v_1=(3,0,-2)$ and $v_2=(2,-1,-5)$ in $R^3(R)$.

Soln:- Since vector $v=(1,-2,k)$ is a linear combination of $v_1=(3,0,-2)$ and $v_2=(2,-1,-5)$; therefore there exist scalars a and b such that

$$v = av_1 + bv_2$$

$$(1,-2,k) = a(3,0,-2)+b(2,-1,-5)$$

$$(1,-2,k) = (3a+2b\ ,\ -b\ ,\ -2a-5b)$$

Equating the corresponding elements,

$$3a+2b=1 \quad , \quad -b=-2, \quad -2a-5b=k$$

$$3a+4=1 \qquad b=2 \qquad -2(-1)-5(2)=k$$

$$a=-1 \qquad\qquad 2-10=k$$

$$\boxed{\therefore\ k=-8.}$$

---

5) Find a condition on a,b,c so that $w=(a,b,c)$ is a linear combination of $u=(1,-3,2)$ and $v=(2,-1,1)$ in R^3 so that $w \in \text{Span}(u,v)$.

Soln:- Let $w = xu+yv$

$$(a,b,c) = x(1,-3,2)+y(2,-1,1)$$

$$x+2y=a\ ;\ -3x-y=b\ ;\ 2x+y=c$$

Let $AX=B$.

Consider $[A:B] = \begin{bmatrix} 1 & 2 & : & a \\ -3 & -1 & : & b \\ 2 & 1 & : & c \end{bmatrix}$

$R_2 \to R_2+3R_1$

$R_3 \to R_3-2R_1$

$$[A:B] = \begin{bmatrix} 1 & 2 & : & a \\ 0 & 5 & : & b+3a \\ 0 & -3 & : & c-2a \end{bmatrix}$$

$$R_2 \rightarrow \frac{R_2}{5}$$

$$[A:B] = \begin{bmatrix} 1 & 2 & : & a \\ 0 & 1 & : & \frac{b+3a}{5} \\ 0 & -3 & : & c-2a \end{bmatrix}$$

$$R_3 \rightarrow R_3 + 3R_2$$

$$[A:B] = \begin{bmatrix} 1 & 2 & : & a \\ 0 & 1 & : & \frac{b+3a}{5} \\ 0 & 0 & : & \frac{5c-a+3b}{5} \end{bmatrix}$$

Rank $[A] = 2$

The system of linear equations will be consistent if rank $[A:B] = 2$

So, $\frac{5c-a+3b}{5} = 0 \Rightarrow a-3b-5c=0$.

i.e. w is linear combination of u and v iff

$a-3b-5c=0$.

6) Express the matrix $A = \begin{bmatrix} 3 & -1 \\ 1 & -2 \end{bmatrix}$ in the vector space of 2×2 matrices as a linear combination of

$B = \begin{bmatrix} 1 & 1 \\ 0 & -1 \end{bmatrix}$, $C = \begin{bmatrix} 1 & 1 \\ -1 & 0 \end{bmatrix}$, $D = \begin{bmatrix} 1 & -1 \\ 0 & 0 \end{bmatrix}$

Soln :- let $A = a_1 B + a_2 C + a_3 D$; $a_1, a_2, a_3 \in R$. $\rightarrow (1)$

$$\begin{bmatrix} 3 & -1 \\ 1 & -2 \end{bmatrix} = a_1 \begin{bmatrix} 1 & 1 \\ 0 & -1 \end{bmatrix} + a_2 \begin{bmatrix} 1 & 1 \\ -1 & 0 \end{bmatrix} + a_3 \begin{bmatrix} 1 & -1 \\ 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 3 & -1 \\ 1 & -2 \end{bmatrix} = \begin{bmatrix} a_1 + a_2 + a_3 & a_1 + a_2 - a_3 \\ -a_2 & -a_1 \end{bmatrix}$$

Equating the corresponding elements,

$$a_1 + a_2 + a_3 = 3 \; ; \; a_1 + a_2 - a_3 = -1 \; ; \; -a_2 = 1 \; ; \; -a_1 = -2$$

Solving the above equations,

$$a_1 = 2, \; a_2 = -1, \; a_3 = 2 .$$

$\therefore$ ① $\Rightarrow$

$$A = 2B - C - 2D .$$

Or

$$\begin{bmatrix} 3 & -1 \\ 1 & 2 \end{bmatrix} = 2\begin{bmatrix} 1 & 1 \\ 0 & -1 \end{bmatrix} - \begin{bmatrix} 1 & 1 \\ -1 & 0 \end{bmatrix} - 2\begin{bmatrix} 1 & -1 \\ 0 & 0 \end{bmatrix}$$

which is the required linear combination of A.

---

7) Express the matrix $\begin{bmatrix} 2 & 0 \\ 4 & -5 \end{bmatrix}$ as a linear combination of the matrices $A = \begin{bmatrix} 0 & -3 \\ 2 & 0 \end{bmatrix}$, $B = \begin{bmatrix} 0 & 0 \\ 2 & 1 \end{bmatrix}$, $C = \begin{bmatrix} 2 & 3 \\ 0 & 5 \end{bmatrix}$.

Soln:- Let $\begin{bmatrix} 2 & 0 \\ 4 & -5 \end{bmatrix} = a_1 \begin{bmatrix} 0 & -3 \\ 2 & 0 \end{bmatrix} + a_2 \begin{bmatrix} 0 & 0 \\ 2 & 1 \end{bmatrix} + a_3 \begin{bmatrix} 2 & 3 \\ 0 & 5 \end{bmatrix}$;

$a_1, a_2, a_3 \in R$.

$$\begin{bmatrix} 2 & 0 \\ 4 & -5 \end{bmatrix} = \begin{bmatrix} 0 + 0 + 2a_3 & -3a_1 + 3a_3 \\ 2a_1 + 2a_2 & a_2 + 5a_3 \end{bmatrix}$$

Equating the corresponding elements

$2 = 2a_3 \rightarrow$ ①

$0 = -3a_1 + 3a_3 \rightarrow$ ②

$4 = 2a_1 + 2a_2 \rightarrow$ ③

$-5 = a_2 + 5a_3 . \rightarrow$ ④

$$2 = 2a_3 \Rightarrow a_3 = 1$$

$$0 = -3a_1 + 3a_3 \Rightarrow a_1 = 1$$

$$4 = 2a_1 + 2a_2 \Rightarrow a_2 = 1$$

But $a_1 = 1, a_2 = 1, a_3 = 1$ do not satisfy eq^n ④

Thus equations ①, ② & ③ have no solution.

Hence, the given matrix cannot be expressed as a linear combination of A, B, C.

Note :-

* Two vectors v_1 and v_2 are linearly dependent iff one of them is a multiple of the other.

Eg:- Determine whether or not the vectors v_1, v_2 are linearly dependent.

i) $v_1 = (1, 3, 9)$ $v_2 = (2, 4, 1)$.

→ No vector is the multiple of the other hence v_1 & v_2 are not linearly dependent.

ii) $v_1 = (3, 4)$ $v_2 = (6, 8)$.

→ $v_2 = 2v_1$ hence v_1 and v_2 are linearly dependent vectors.

---

8) Determine whether the vectors $v_1 = (1, 2, 3)$ $v_2 = (3, 1, 7)$ and $v_3 = (2, 5, 8)$ are linearly dependent or linearly independent.

Soln:- $x v_1 + y v_2 + z v_3 = 0$

$x(1, 2, 3) + y(3, 1, 7) + z(2, 5, 8) = (0, 0, 0)$

$x + 3y + 2z = 0$; $2x + y + 5z = 0$; $3x + 7y + 8z = 0$.

Let $AX = 0$ where $A = \begin{bmatrix} 1 & 3 & 2 \\ 2 & 1 & 5 \\ 3 & 7 & 8 \end{bmatrix}$

$R_2 \to R_2 - 2R_1$
$R_3 \to R_3 - 3R_1$

$A = \begin{bmatrix} 1 & 3 & 2 \\ 0 & -5 & 1 \\ 0 & -2 & 2 \end{bmatrix}$ $R_2 \to -R_2$
$R_3 \to -\frac{R_3}{2}$

$A = \begin{bmatrix} 1 & 3 & 2 \\ 0 & 5 & -1 \\ 0 & 1 & -1 \end{bmatrix}$ $R_2 \leftrightarrow R_3$

$$A = \begin{bmatrix} 1 & 3 & 2 \\ 0 & 1 & -1 \\ 0 & 5 & -1 \end{bmatrix}$$

$R_3 \rightarrow R_3 - 5R_2$

$$A = \begin{bmatrix} 1 & 3 & 2 \\ 0 & 1 & -1 \\ 0 & 0 & 4 \end{bmatrix}$$

Rank [A] = 3 = Number of unknowns.

$\Rightarrow$ Homogeneous system of linear equations possesses unique solution.

i.e $x=0,\ y=0,\ z=0$.

So vectors v_1, v_2, v_3 are linearly independent.

9) Determine whether the vectors $v_1 = (1,9,3)$, $v_2 = (2,5,4)$, $v_3 = (0,0,0)$ are linearly dependent or linearly independent.

Soln:- vector v_1, v_2, v_3 are linearly dependent as linearly independent set of vectors can not contain zero vector.

---

10) Determine whether the vectors $v_1 = (1,4,9)$, $v_2 = (3,1,4)$, $v_3 = (9,3,12)$ are linearly dependent or linearly independent.

Soln:- $x v_1 + y v_2 + z v_3 = 0$.

$x(1,4,9) + y(3,1,4) + z(9,3,12) = (0,0,0)$

$x + 3y + 9z = 0$; $4x + y + 3z = 0$; $9x + 4y + 12z = 0$.

Consider $AX = 0$. where

$$A = \begin{bmatrix} 1 & 3 & 9 \\ 4 & 1 & 3 \\ 9 & 4 & 12 \end{bmatrix}$$

$R_2 \rightarrow R_2 - 4R_1$

$R_3 \rightarrow R_3 - 9R_1$

$$= \begin{bmatrix} 1 & 3 & 9 \\ 0 & -11 & -33 \\ 0 & -23 & -69 \end{bmatrix} \quad \begin{matrix} R_2 \to -\frac{R_2}{11} \\ R_3 \to -\frac{R_3}{29} \end{matrix}$$

$$= \begin{bmatrix} 1 & 3 & 9 \\ 0 & 1 & 3 \\ 0 & 1 & 3 \end{bmatrix} \quad R_3 \to R_3 - R_2$$

$$= \begin{bmatrix} 1 & 3 & 9 \\ 0 & 1 & 3 \\ 0 & 0 & 0 \end{bmatrix}$$

Rank $[A] = 2 < 3$ (Number of unknowns)

So one unknown is constant and hence not always zero.

So the vectors are linearly dependent.

[OR]

$v_3 = 3v_2$

Vectors v_1, v_2, v_3 are linearly dependent.

---

11) If u, v, w are linearly independent vectors in $V(F)$, where F is the field of complex numbers, then $\{u+v, v+w, w+u\}$ is a linearly independent set of vectors.

Soln:- Let $a(u+v) + b(v+w) + c(w+u) = 0$, where $a, b, c \in F$

$$(a+c)u + (a+b)v + (b+c)w = 0 \rightarrow (1)$$

Since u, v, w are linearly independent.

$\therefore$ (1) $\Rightarrow$ $a+c=0$

$a+b=0$

$b+c=0$.

$\therefore$ $a=0, b=0, c=0$.

$\therefore$ $\{u+v, v+w, w+u\}$ is a linearly independent set of vectors.

12) Let V be the vector space of all 2x3 matrices over R. S.T the matrices $A=\begin{bmatrix} 2 & 1 & -1 \\ 3 & -2 & 4 \end{bmatrix}$ $B=\begin{bmatrix} 1 & 1 & -3 \\ -2 & 0 & 5 \end{bmatrix}$ and $C=\begin{bmatrix} 4 & -1 & 2 \\ 1 & -2 & 3 \end{bmatrix}$ form a linearly independent set.

Soln:- Let $aA+bB+cC=0$ where $a,b,c \in R$.

$$\Rightarrow a\begin{bmatrix} 2 & 1 & -1 \\ 3 & -2 & 4 \end{bmatrix}+b\begin{bmatrix} 1 & 1 & -3 \\ -2 & 0 & 5 \end{bmatrix}+c\begin{bmatrix} 4 & -1 & 2 \\ 1 & -2 & 3 \end{bmatrix}=\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 2a+b+4c & a+b-c & -a-3b+2c \\ 3a-2b+c & -2a+0b-2c & 4a+5b+3c \end{bmatrix}=\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

By equating the corresponding elements

$$\left.\begin{aligned} 2a+b+4c&=0 \\ a+b-c&=0 \\ -a-3b+2c&=0 \end{aligned}\right\} \rightarrow ① \qquad \left.\begin{aligned} 3a-2b+c&=0 \\ -2a-2c&=0 \\ 4a+5b+3c&=0 \end{aligned}\right\} \rightarrow ②$$

On solving the system ① of equations, we observe that the only solution is $a=0, b=0, c=0$. Clearly this solution also satisfies the system (2) of equations.

Hence the given set of matrices is linearly independent.

---

13) Show that the vectors $(1,1,2,4),(2,-1,-5,2),(1,-1,-4,0)$ and $(2,1,1,6)$ are linearly dependent.

14) Determine whether or not each of the following forms a basis, $x_1=(1,2,3)$, $x_2=(3,-5,6)$ in R^3.

Soln:- Basis in R^3 must contain exactly 3 elements therefore x_1, x_2 does not form a basis in R^3.

15) Determine whether or not each of the following form a basis $x_1 = (1,2,9)$, $x_2 = (2,-3,4)$, $x_3 = (1,3,7)$, $x_4 = (2,4,8)$ in R^3.

Soln:- Basis in R^3 must contain exactly 3 elements.

$\therefore$ x_1, x_2, x_3, x_4 does not form a basis in R^3.

---

16) Determine whether or not each of the following forms a basis, $x_1 = (2,2,1)$, $x_2 = (1,3,7)$, $x_3 = (1,2,2)$ in R^3.

Soln:- Three vectors in R^3 form a basis iff they are linearly independent.

$$x(2,2,1) + y(1,3,7) + z(1,2,2) = (0,0,0)$$

$$2x+y+z=0 ; \quad 2x+3y+2z=0 ; \quad x+7y+2z=0.$$

Let $AX=0$ where

$$A = \begin{bmatrix} 2 & 1 & 1 \\ 2 & 3 & 2 \\ 1 & 7 & 2 \end{bmatrix} \quad R_1 \leftrightarrow R_3$$

$$A = \begin{bmatrix} 1 & 7 & 2 \\ 2 & 3 & 2 \\ 2 & 1 & 1 \end{bmatrix} \quad \begin{matrix} R_2 \to R_2 - 2R_1 \\ R_3 \to R_3 - 2R_1 \end{matrix}$$

$$A = \begin{bmatrix} 1 & 7 & 2 \\ 0 & -11 & -2 \\ 0 & -13 & -3 \end{bmatrix} \quad \begin{matrix} R_2 \to -R_2 \\ R_3 \to -R_3 \end{matrix}$$

$$A = \begin{bmatrix} 1 & 7 & 2 \\ 0 & 11 & 2 \\ 0 & 13 & 3 \end{bmatrix}$$

$$R_3 \to R_3 - R_2$$

$$A = \begin{bmatrix} 1 & 7 & 2 \\ 0 & 11 & 2 \\ 0 & 2 & 1 \end{bmatrix}$$

$R_3 \to \frac{R_3}{2}$

$$A = \begin{bmatrix} 1 & 7 & 2 \\ 0 & 11 & 2 \\ 0 & 1 & 1/2 \end{bmatrix}_{R_3 \leftrightarrow R_2} = \begin{bmatrix} 1 & 7 & 2 \\ 0 & 1 & 1/2 \\ 0 & 11 & 2 \end{bmatrix}$$

$R_3 \to R_3 - 11R_2$

$$A = \begin{bmatrix} 1 & 7 & 2 \\ 0 & 1 & 1/2 \\ 0 & 0 & -7/2 \end{bmatrix}$$

Rank [A] = 3 = Number of unknowns. So unique solution and solution is

$x = 0, \ y = 0, \ z = 0$.

Hence vectors x_1, x_2, x_3 are linearly independent and hence form a basis.

---

17) Let W be the subspace of $\mathbb{R}^5$, spanned by $x_1 = (1,2,-1,3,4)$, $x_2 = (2,4,-2,6,8)$, $x_3 = (1,3,2,2,6)$, $x_4 = (1,4,5,1,8)$, $x_5 = (2,7,3,3,9)$.

Find a subset of vectors which forms a basis of W.

Soln:-

$$A = \begin{bmatrix} 1 & 2 & -1 & 3 & 4 \\ 2 & 4 & -2 & 6 & 8 \\ 1 & 3 & 2 & 2 & 6 \\ 1 & 4 & 5 & 1 & 8 \\ 2 & 7 & 3 & 3 & 9 \end{bmatrix}$$

$R_2 \to R_2 - 2R_1$

$R_3 \to R_3 - R_1$

$R_4 \to R_4 - R_1$

$R_5 \to R_5 - 2R_1$

$$A=\begin{bmatrix}1 & 2 & -1 & 3 & 4\\ 0 & 0 & 0 & 0 & 0\\ 0 & 1 & 3 & -1 & 2\\ 0 & 2 & 6 & -2 & 4\\ 0 & 3 & 5 & -3 & 1\end{bmatrix}$$

$R_4 \to R_4 - 2R_3,\ R_5 \to R_5 - 3R_3$

$$A=\begin{bmatrix}1 & 2 & -1 & 3 & 4\\ 0 & 0 & 0 & 0 & 0\\ 0 & 1 & 3 & -1 & 2\\ 0 & 0 & 0 & 0 & 0\\ 0 & 0 & -4 & 0 & -5\end{bmatrix}$$

Number of non zero rows = 3

$\therefore$ dim $W = 3$ and $\{x_1, x_3, x_5\}$ forms a basis in W.

---

18) V is a vector space of polynomials over R. Find a basis and dimension of the subspace W of V, spanned by the polynomials.

$x_1 = t^3 - 2t^2 + 4t + 1$, $x_2 = 2t^3 - 3t^2 + 9t - 1$

$x_3 = t^3 + 6t - 5$, $x_4 = 2t^3 - 5t^2 + 7t + 5$.

Soln:- $A=\begin{bmatrix}1 & -2 & 4 & 1\\ 2 & -3 & 9 & -1\\ 1 & 0 & 6 & -5\\ 2 & -5 & 7 & 5\end{bmatrix} = \begin{bmatrix}1 & -2 & 4 & 1\\ 0 & 1 & 1 & -3\\ 0 & 2 & 2 & -6\\ 0 & -1 & -1 & 3\end{bmatrix}$

$R_2 \to R_2 - 2R_1$
$R_3 \to R_3 - R_1$
$R_4 \to R_4 - 2R_1$

$R_3 \to R_3 - 2R_2$
$R_4 \to R_4 + R_2$

$$A=\begin{bmatrix}1 & -2 & 4 & 1\\ 0 & 1 & 1 & -3\\ 0 & 0 & 0 & 0\\ 0 & 0 & 0 & 0\end{bmatrix}$$

Non Zero rows of Echelon matrix forms a basis.

Number of non Zero rows = 2

$\therefore$ dim $W = 2$ and $\{x_1, x_2\}$ forms a basis in W.

Linear Transformations :-

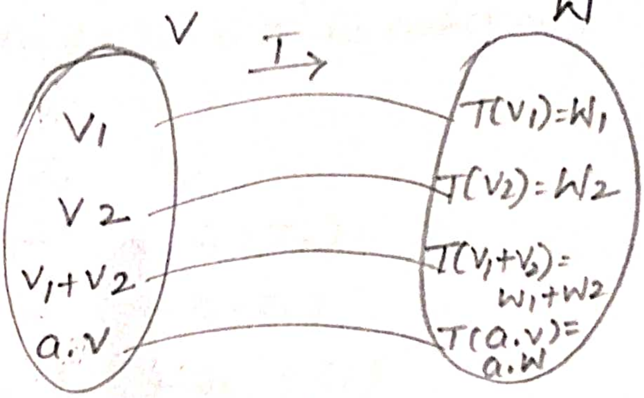
Let V and W be any two subspaces over the field F. A mapping T from V to W is called a linear transformation if,

i) $v_1, v_2 \in V$

$T(v_1+v_2) = T(v_1) + T(v_2)$

ii) $\forall a \in F$ and $\forall v \in V$

$T(a.v) = a.T(v)$



Problems :-

1) Show that the function $T: R^2 \to R^3$ given by $T(x,y) = (x+y, x-y, y)$ is a linear transformation.

Soln :- Let $u=(x_1,y_1)$, $v=(x_2,y_2)$ be two vectors belonging to R^2.

$$\therefore T(u+v) = T(x_1+x_2, y_1+y_2)$$

$$= (x_1+x_2+y_1+y_2, x_1+x_2-y_1-y_2, y_1+y_2)$$

$$= \{(x_1+y_1)+(x_2+y_2), (x_1-y_1)+(x_2-y_2), (y_1+y_2)\}$$

$$= (x_1+y_1, x_1-y_1, y_1) + (x_2+y_2, x_2-y_2, y_2)$$

$$= T(u) + T(v)$$

Also for $a \in R$ and $u \in R^2$, we have

$$T(au) = T(ax_1, ay_1)$$

$$= (ax_1+ay_1, ax_1-ay_1, ay_1)$$

$$= a(x_1+y_1 \quad x_1-y_1 \quad y_1)$$

$$= a\,T(u)$$

$\therefore T$ is a linear transformation.

Q) Which of the following functions are linear transformations?

i) $T: R^3 \to R^3$ defined by $T(x,y,z) = (y, -x, -z)$

ii) $T: R^3 \to R^2$ defined by $T(x,y,z) = (2x-3y, 7y+2z)$.

Soln:- i) Let $u = (x_1, y_1, z_1)$, $v = (x_2, y_2, z_2) \in R^3$ be arbitrary.

$$T(u) = T(x_1, y_1, z_1) = (y_1, -x_1, -z_1)$$

$$T(v) = T(x_2, y_2, z_2) = (y_2, -x_2, -z_2)$$

Now, $T(u+v) = T(x_1+x_2, y_1+y_2, z_1+z_2)$

$$= (y_1+y_2, -x_1-x_2, -z_1-z_2)$$

$$= (y_1, -x_1, -z_1) + (y_2, -x_2, -z_2)$$

$$= T(u) + T(v)$$

Also, for any scalar $a \in R$,

$$T(au) = T(ax_1, ay_1, az_1)$$

$$= (ay_1, -ax_1, -az_1)$$

$$= a(y_1, -x_1, -z_1)$$

$$= aT(u)$$

Hence, T is a linear transformation.

ii) Let $u = (x_1, y_1, z_1)$ and $v = (x_2, y_2, z_2) \in R^3$ be arbitrary.

$$T(u) = T(x_1, y_1, z_1) = (2x_1 - 3y_1, 7y_1 + 2z_1)$$

$$T(v) = T(x_2, y_2, z_2) = (2x_2 - 3y_2, 7y_2 + 2z_2)$$

$\therefore T(u+v) = T(x_1+x_2, y_1+y_2, z_1+z_2)$

$$= (2x_1 + 2x_2 - 3y_1 - 3y_2, 7y_1 + 7y_2 + 2z_1 + 2z_2)$$

$$= \{(2x_1 - 3y_1) + (2x_2 - 3y_2), (7y_1 + 2z_1) + (7y_2 + 2z_2)\}$$

$$= (2x_1 - 3y_1, 7y_1 + 2z_1) + (2x_2 - 3y_2, 7y_2 + 2z_2)$$

$$= T(u) + T(v)$$

Also, for any scalar $a \in R$

$$T(au) = T(ax_1, ay_1, az_1)$$

$$= (2ax_1 - 3ay_1, 7ay_1 + 2az_1)$$

$= a(2x_1 - 3y_1, 7y_1 + 2z_1)$

$= aT(u)$.

Hence, T is a linear transformation.

3) Let $T: V(R) \to V_2(R)$ be a mapping $f(x) = (3x, 5x)$. Show that f is a linear transformation.

Soln:– i) $f(x_1 + x_2) = (3(x_1 + x_2), 5(x_1 + x_2))$

$= (3x_1 + 3x_2, 5x_1 + 5x_2)$

$= (3x_1, 5x_1) + (3x_2, 5x_2)$

$= f(x_1) + f(x_2)$

ii) $f(a \cdot x) = (3(a \cdot x), 5(a \cdot x))$

$= (3ax, 5ax)$

$= a(3x, 5x)$

$= a \cdot f(x)$.

Hence f is a linear transformation.

4) Show that the translation mapping $f: V_2(R) \to V_2(R)$ defined by $f(x, y) = (x+6, y+2)$ is not linear.

Soln:– $f(x_1 + x_2, y_1 + y_2) = (x_1 + x_2 + 6, y_1 + y_2 + 2) \to (1)$

$f(x_1, y_1) + f(x_2, y_2) = (x_1 + 6, y_1 + 2) + (x_2 + 6, y_2 + 2)$

$= (x_1 + x_2 + 12, y_1 + y_2 + 4) \to (2)$

Equation (1) $\neq$ Equation (2).

Hence f is not a linear transformation.

* Kernel of T and Image of T:

Let T be a linear transformation from V to W

Kernel of T or $Ker(T) = \{v \in V : T(v) = 0, 0 \in W\}$

Remark :- Kernel of T is also known as null space of T.

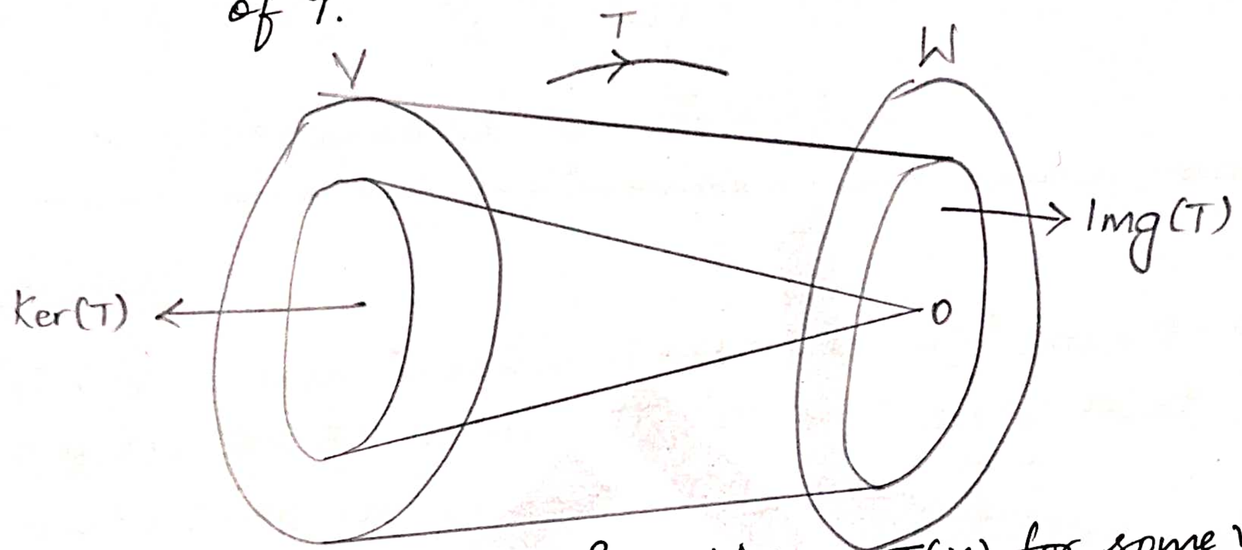


Image of T or $Img(T) = \{w \in W ; w = T(v) \text{ for some } v \in V\}$

Algebra of Linear Transformations :-

* Sum of Linear transformations:-

Let $T_1 : U \to V$ and $T_2 : U \to V$ be two transformations where U and V are two vectors spaces over the field F. We define the sum of T_1 and T_2 by $T_1 + T_2 : U \to V$, such that $(T_1 + T_2)u = T_1(u) + T_2(u), u \in U$.

* Scalar multiplication of Linear transformations:-

If $a \in F$, the function aT, i.e product of a linear transformation $T : U \to V$ with a, and defined by $(aT) : U \to V$, such that $(aT)u = a(T(u))$ for all $u \in U$ is a linear transformation from U into V.

* Composition (Product) of two Linear Transformations:

Let U, V, W be three vector spaces over a field F. We define $T_2 T_1$, called the composition or product of T_2 and T_1, by $(T_2 T_1)(u) = T_2(T_1(u))$.

$T_2 T_1$ is also denoted by $(T_2 \circ T_1)$.

Problems:-

1) Let the linear transformations $T_1 : R^3 \to R^2$ such that $T_1(x, y, z) = (4x, 3y - 2z)$ and $T_2 : R^2 \to R^2$ such that $T_2(x, y) = (-2x, y)$. Compute $T_1 T_2$ and $T_2 T_1$.

Soln:- As the range of T_2 is not contained in the domain of T_1

$\therefore$ $T_1 T_2$ is not defined.

Now, $T_2 T_1$ is defined as the range of T_1 is contained in the domain of T_2.

$$T_2 T_1 (x, y, z) = T_2 (T_1 (x, y, z))$$
$$= T_2 (4x, 3y - 2z)$$
$$= (-8x, 3y - 2z).$$

---

2) Illustrate with the help of an example that there exist linear transformations $T_1 : R^2 \to R^2$ and $T_2 : R^2 \to R^2$ such that $T_2 T_1 = 0$ but $T_1 T_2 \neq 0$.

Soln:- Let us define $T_1 : R^2 \to R^2$ by
$$T_1(x, y) = (0, 4x)$$
& $T_2 : R^2 \to R^2$ by $T_2(x, y) = (x, 0)$

$\therefore$ $T_1 T_2$ and $T_2 T_1$ both are defined.

$(T_2T_1)(x,y) = T_2(T_1(x,y)) = T_2(0,4x)$
$= (0,0) = 0(x,y)$

$T_2T_1 = 0$.

Also $(T_1T_2)(x,y) = T_1(T_2(x,y)) = T_1(x,0)$
$= (0,4x) \neq 0(x,y)$

$T_1T_2 \neq 0$.

Matrix of a Linear transformation:—

Let $T: U \to V$ be the linear transformation, where U and V are vector spaces over field F.

Let $B = \{u_1, u_2 ---- u_n\}$ and $B' = \{v_1, v_2 --- v_m\}$ be Ordered basis for the finite dimensional vector spaces U and V respectively.

Since $T(u_1), T(u_2) ---- T(u_n) \in V$ and $\{v_1, v_2 --- v_m\}$ spans V, each $T(u_i)$ can be expressed as a linear combination of the vectors $v_1, v_2 --- v_m$.

Let $T(u_1) = a_{11}v_1 + a_{21}v_2 + ---- + a_{m1}v_m$
$T(u_2) = a_{12}v_1 + a_{22}v_2 + ---- + a_{m2}v_m$

$- - - - - - - - -$

$T(u_n) = a_{1n}v_1 + a_{2n}v_2 + ---- + a_{mn}v_m$ where $a_{ij} \in F$.

The coefficient matrix of this system of equations is

$$\begin{bmatrix} a_{11} & a_{21} & a_{31} & ---- & a_{m1} \\ a_{12} & a_{22} & a_{32} & ---- & a_{m2} \\ - & - & - & ---- & - \\ a_{1n} & a_{2n} & a_{3n} & --- & a_{mn} \end{bmatrix}$$

The transpose of this matrix is a matrix representation of T, called matrix of T with repect to ordered basis B and B' (or matrix associated with T w.r.to B and B'] It is denoted by [T:B, B'] and is given by

$$[T:B,B'] = \begin{bmatrix} a_{11} & a_{12} & - & - & a_{1n} \\ a_{21} & a_{22} & - & - & a_{2n} \\ \vdots & \vdots & - & - & - \\ a_{m1} & a_{m2} & - & - & a_{mn} \end{bmatrix}$$

Problems:-

1) Find the matrix representing the transformation $T: R^2 \to R^3$ given by $T(x_1, x_2) = (3x_1 - x_2, 2x_1 + 4x_2, 5x_1 - 6x_2)$ relative to the standard basis of R^2 and R^3.

Soln:- The ordered standard basis of R^2 is $B = \{(1,0), (0,1)\}$ and that of R^3 is

$B' = \{(1,0,0), (0,1,0), (0,0,1)\}$.

$\therefore T(x_1, x_2) = (3x_1 - x_2, 2x_1 + 4x_2, 5x_1 - 6x_2)$

$T(1, 0) = (3,2,5)$ & $T(0,1) = (-1,4,-6)$

$T(1,0) = (3,2,5) = 3(100) + 2(010) + 5(001)$

$T(0,1) = (-1,4,-6) = -1(100) + 4(010) + (-6)(001)$

Hence, matrix of the transformation is $\begin{bmatrix} 3 & -1 \\ 2 & 4 \\ 5 & -6 \end{bmatrix}$

---

2) Find the matrix representing the transformation $T: R^3 \to R^4$ defined by $T(x,y,z) = (x+y+z, 2x+z, 2y-z, 6y)$ relative to the standard basis of R^3 and R^4.

Sol^n:- We know that ordered standard basis of R^3 is
$B = \{(1,0,0), (0,1,0), (0,0,1)\}$ and
R^4 is $B' = \{(1,0,0,0), (0,1,0,0), (0,0,1,0), (0,0,0,1)\}$
Since $T(x,y,z) = (x+y+z,\ 2x+z,\ 2y-z,\ 6y)$

$$T(1,0,0) = (1,2,0,0)$$
$$T(0,1,0) = (1,0,2,6)$$
$$T(0,0,1) = (1,1,-1,0)$$

Let $e_1 = (1,0,0,0)$ $e_2 = (0,1,0,0)$ $e_3 = (0,0,1,0)$ $e_4 = (0,0,0,1)$

$$\therefore \left.\begin{aligned} T(1,0,0) &= (1,2,0,0) = 1e_1 + 2e_2 + 0e_3 + 0e_4 \\ T(0,1,0) &= (1,0,2,6) = 1e_1 + 0e_2 + 2e_3 + 6e_4 \\ T(0,0,1) &= (1,1,-1,0) = 1e_1 + 1e_2 - 1e_3 + 0e_4 \end{aligned}\right\} \rightarrow (1)$$

Matrix of T relative to B and B' is transpose of matrix of coefficients in above system of eq^n (1)

i.e, $$[T:B,B'] = \begin{bmatrix} 1 & 1 & 1 \\ 2 & 0 & 1 \\ 0 & 2 & -1 \\ 0 & 6 & 0 \end{bmatrix}.$$

---

3) Find the linear map $T: R^2 \rightarrow R^3$ whose matrix is
$A = \begin{bmatrix} 1 & -1 \\ -2 & 3 \\ 0 & 1 \end{bmatrix}$, relative to the ordered basis $B = \{(1,1)\ (0,2)\}$ and
$B' = \{(0,1,1), (1,0,1), (1,1,0)\}$ for R^3.

Sol^n:- Here $[T:B,B'] = \begin{bmatrix} 1 & -1 \\ -2 & 3 \\ 0 & 1 \end{bmatrix}$

$T(1,1)$ = Linear Combination of vectors of B' using scalars of first column of $[T:B,B']$
$$= 1(0,1,1) - 2(1,0,1) + 0(1,1,0) = (-2,1,-1)$$
$|||^{ly}$ $T(0,2) = -1(0,1,1) + 3(1,0,1) + 1(1,1,0) = (4,0,2)$

Let $(x,y) \in R^2$ be arbitrary.

$$(x,y) = a(1,1) + b(0,2)$$

To find a and b :—

$$(x,y) = (a, a+2b) \Rightarrow a = x$$

and $a+2b = y \Rightarrow b = \frac{y-x}{2}$

Using these values,

$$(x,y) = x(1,1) + \frac{y-x}{2}(0,2)$$

Applying T on both sides,

$$T(x,y) = x\,T(1,1) + \left(\frac{y-x}{2}\right)T(0,2)$$

$$= x(-2,1,-1) + \left(\frac{y-x}{2}\right)(4,0,2)$$

$$= \left[-2x + \left(\frac{y-x}{2}\right)4,\ x+0,\ -x + \left(\frac{y-x}{2}\right)2\right]$$

$= (2y-4x,\ x,\ y-2x)$ Which is the required transformation.

Change of coordinates or change of Basis Matrix or Transition Matrix :—

Let $B_1 = \{u_1, u_2, \ldots u_n\}$ be a basis of n-dimensional vector space V and $B_2 = \{v_1, v_2, \ldots v_n\}$ be another basis of V. Then, each element in B_2 can be expressed as a linear combination of the vectors in basis B_1.

Let $v_1 = a_{11}u_1 + a_{12}u_2 + \cdots + a_{1n}u_n$

$v_2 = a_{21}u_1 + a_{22}u_2 + \cdots + a_{2n}u_n$

$\vdots$

$v_n = a_{n1}u_1 + a_{n2}u_2 + \cdots + a_{nn}u_n$

$$\Rightarrow \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ - & - & - & - \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \\ \vdots \\ u_n \end{bmatrix}$$

Then the transpose of the matrix of coefficients in above system of equations is said to be the transition matrix or change of basis matrix from the old basis B_1 to the new basis B_2. It is denoted by P and written as

$$P = \begin{bmatrix} a_{11} & a_{21} & \cdots & a_{n1} \\ a_{12} & a_{22} & \cdots & a_{n2} \\ - & - & - & - \\ a_{1n} & a_{2n} & \cdots & a_{nn} \end{bmatrix}$$

Since the vectors in B_2 are linearly independent, so the matrix P is invertible. Its inverse P^{-1} is the transition matrix from the basis B_2 to the basis B_1.

Problems:-

1) Consider the following basis of R^2:

$E = \{e_1, e_2\} = \{(1\,0), (0\,1)\}$ and $S = \{u_1, u_2\} = \{(1,3), (1,4)\}$

(a) Find the change-of-basis matrix P from the usual basis E to S.

(b) Find the change-of-basis matrix Q from S back to E.

(c) Find the coordinate vector $[v]$ of $v = (5,-3)$ relative to S.

(a) Since E is the usual basis of R^2.

$\therefore \; P = \begin{bmatrix} 1 & 1 \\ 3 & 4 \end{bmatrix}$

(b) Find P^{-1}, $P^{-1} = \begin{bmatrix} 4 & -1 \\ -3 & 1 \end{bmatrix}$

(c) Let $[v]_S = P^{-1}[v]_E$.

We have $[v]_E = [5, -3]^T$

$[v]_S = P^{-1}[v]_E = \begin{bmatrix} 4 & -1 \\ -3 & 1 \end{bmatrix}\begin{bmatrix} 5 \\ -3 \end{bmatrix} = \begin{bmatrix} 23 \\ -18 \end{bmatrix}$.

---

2) The vectors $u_1 = (1,2,0)$, $u_2 = (1,3,2)$, $u_3 = (0,1,3)$ form a basis S of R^3. Find:

a) The change of basis matrix P from the usual basis $E = \{e_1, e_2, e_3\}$ of R^3 to the basis S.

b) The change of basis matrix Q from the above basis S back to the usual basis E of R^3.

Soln:-

(a) Since E is a usual basis of R^3.

$P = \begin{bmatrix} 1 & 1 & 0 \\ 2 & 3 & 1 \\ 0 & 2 & 3 \end{bmatrix}$

(b) Find P^{-1}, $P^{-1} = \begin{bmatrix} 7 & -3 & 1 \\ -6 & 3 & -1 \\ 4 & -2 & 1 \end{bmatrix}$

3) Find the co-ordinates of vector $(1,1,1)$ relative to basis $(1,1,2), (2,2,1), (1,2,2)$.

Solⁿ:- Let $(1,1,1) = a(1,1,2) + b(2,2,1) + c(1,2,2)$

$$a+2b+c=1 \rightarrow (1)$$
$$a+2b+2c=1 \rightarrow (2)$$
$$2a+b+2c=1 \rightarrow (3)$$

Subtracting (1) from (2), we have $c=0$

Putting $c=0$ in (2) and (3), we get

$$a+2b=1 \rightarrow (4)$$
$$2a+b=1 \rightarrow (5)$$

Solving (4) and (5), we get $a=b=\frac{1}{3}$.

$\therefore$ Coordinates of $(1,1,1)$ relative to given basis are $\left(\frac{1}{3}, \frac{1}{3}, 0\right)$.

---

4) Find the co-ordinates of the following vectors relative to the basis $\gamma_1 = (1,1,2)$, $\gamma_2 = (2,2,1)$, $\gamma_3 = (1,2,2)$.

(i) $(1,0,1)$ solⁿ:- $\left(\frac{4}{3}, \frac{1}{3}, -1\right)$

(ii) $(1,1,0)$ solⁿ:- $\left(-\frac{1}{3}, \frac{2}{3}, 0\right)$

Rank and nullity of a linear Operator:—

Rank of T:—

Dimension of the Img(T) is known as rank of T.

Nullity of T:—

Dimension of the Ker(T) is known as nullity of T.

Rank-nullity Theorem:—

Let V and W be vector spaces over the field F and let T be a linear transformation from V to W.
If V is a finite dimensional then

$$\text{rank}(T) + \text{nullity of }(T) = \dim(V).$$

Problems:—

1) For the linear transformation $T: R^2 \to R^3$ such that $T(x_1, x_2) = (x_1 - x_2,\ x_2 - x_1,\ -x_1)$, find a basis and dimension of its range space and its null space. Also verify that $\text{rank}(T) + \text{nullity}(T) = \dim R^2$.

Soln:- i) To find ~~all~~ null space of T and its dimension:

By defn, null space $= N(T) = \{u \in R^2 : T(u) = 0 \in R^3\}$

Let $u = (x_1, x_2) \in R^2$ be an arbitrary element of null space.

$$T(u) = 0$$

$$T(x_1, x_2) = 0$$

$$(x_1 - x_2,\ x_2 - x_1,\ -x_1) = 0$$

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$\Rightarrow \quad x_1 - x_2 = 0$

$x_2 - x_1 = 0$

$-x_1 = 0$

On solving the above equations, we get

$x_1 = 0, \ x_2 = 0.$

Thus, the only member of null space is a zero vector $\in R^2$

i.e $N(T) = \{0\}$.

$\therefore$ Nullity of $T = \dim(N(T)) = 0$.

ii) To find range space of T and its dimension:

The range space consists of ordered triples $(x_1 - x_2, x_2 - x_1, -x_1)$ for $(x_1, x_2) \in R^2$ such that

$$T(x_1, x_2) = (x_1 - x_2, x_2 - x_1, -x_1) \rightarrow (1)$$

Let v be any element of $R(T)$.

Thus there exist $(x_1, x_2) \in R^2$ such that $v = T(x_1, x_2)$

$\Rightarrow \quad (x_1, x_2) = x_1(1,0) + x_2(0,1)$

$= x_1 e_1 + x_2 e_2$ where $e_1 = (1,0), e_2 = (0,1)$

$T(x_1, x_2) = T(x_1 e_1 + x_2 e_2)$

$= x_1 T(e_1) + x_2 T(e_2) \longrightarrow (2)$

$T(e_1) = T(1,0) = (1-0, 0-1, -1) = (1, -1, -1)$

$T(e_2) = T(0,1) = (0-1, 1-0, 0) = (-1, 1, 0)$

Putting the values of $T(e_1), T(e_2)$ in (2)

$T(x_1, x_2) = x_1(1, -1, -1) + x_2(-1, 1, 0)$

$v = x_1(1, -1, -1) + x_2(-1, 1, 0) \rightarrow (3)$

Since $v \in R(T)$ is arbitrary,

(iii) Let $a(1, -1, -1) + b(-1, 1, 0) = 0$ for $a, b \in F$

$\Rightarrow (a-b, -a+b, -a) = (0,0,0)$

$\Rightarrow a-b=0, b-a=0, -a=0$

$\Rightarrow a=b=0.$

Thus $\{(1,-1,-1),(-1,1,0)\}$ is linearly independent, a

$\therefore$ It is a basis set of $R(T)$.

$\therefore$ Dimension of $R(T) = 2$

Also dimension of $R^2 = 2$

$\therefore$ Rank (T) + nullity $(T) = 2+0 = 2 = \dim R^2$.

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2) Find a linear transformation $T: R^3 \to R^3$ whose rangespace is spanned by the vectors $(1,2,3),(4,5,6)$

Soln:- We know that $\{e_1, e_2, e_3\}$ is a standard basis of R^3, where $e_1 = (1,0,0)$, $e_2 = (0,1,0)$, $e_3 = (0,0,1)$

Since R^3 is a three dimensional vector space and $\{e_1, e_2, e_3\}$ is a basis of R^3, there exist a unique linear transformation T such that

$T(e_1) = (1,2,3)$

$T(e_2) = (4,5,6)$

$T(e_3) = (0,0,0)$ (Assuming $T(0,0,1) = (0,0,0)$).

Also $R(T)$ is spanned by $\{(1,2,3),(4,5,6)\}$

i.e by $\{(1,2,3),(4,5,6),(0,0,0)\}$ or by $\{T(e_1), T(e_2), T(e_3)\}$

Now, for each $(x_1, x_2, x_3) \in R^3$,

$$(x_1, x_2, x_3) = x_1(1,0,0) + x_2(0,1,0) + x_3(0,0,1)$$
$$= x_1 e_1 + x_2 e_2 + x_3 e_3$$

$$T(x_1,x_2,x_3) = x_1T(e_1) + x_2T(e_2) + x_3T(e_3)$$
$$= x_1(1,2,3) + x_2(4,5,6) + x_3(0,0,0)$$
$$= (x_1+4x_2,\ 2x_1+5x_2,\ 3x_1+6x_2)$$

Which is the required linear transformation.

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3) Verify the Rank-nullity theorem for the $T: R^3 \to R^3$ defined by $T(x,y,z) = (x+2y-z,\ y+z,\ x+y-2z)$

Soln:- To find the basis of nullity of T

Let $v = (x,y,z) \in R^3$ such that

Nullity of $T = \{v \in V \mid T(v) = 0\}$

$$T(x,y,z) = 0.$$
$$(x+2y-z,\ y+z,\ x+y-2z) = (0,0,0).$$
$$\Rightarrow \quad x+2y-z = 0 \Rightarrow x = 3z$$
$$y+z = 0 \Rightarrow y = -z.$$
$$x+y-2z = 0$$

On solving above equations we get

$$x = 3z, \quad y = -z$$
$$\therefore \{(x,y,z)\} = \{3z, -z, z\} = (z\{3,-1,1\})$$

Thus $\{(3,-1,1)\}$ is a basis of nullity of T

& Nullity $(T) = 1$.

To find the basis of Range of T.

As $\{(1,0,0), (0,10), (0,0,1)\}$ generates R^3

$\Rightarrow \{T(1,0,0), T(0,1,0), T(0,0,1)\}$ generates range of T

$\Rightarrow \{(1,0,1), (2,1,1), (-1,1,-2)\}$ generates range of T

To find the basis of range,
Consider a matrix.

$$X = \begin{bmatrix} 1 & 0 & 1 \\ 2 & 1 & 1 \\ -1 & 1 & -2 \end{bmatrix} \Rightarrow \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & -1 \\ 0 & 1 & -1 \end{bmatrix} \Rightarrow \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{bmatrix}$$

$R_2 \to R_2 - 2R_1$, $R_3 \to R_3 + R_2$; $R_3 \to R_3 - R_2$

Thus $\{(1,0,1), (0,1,-1)\}$ form a basis of Range of T & $\dim(\text{Range of } T) = 2$.

$\therefore$ Rank (T) + Nullity (T)

$= 2 + 1$

$= 3 = \dim(R^3)$.

Hence rank-Nullity theorem verified.

4) Verify the Rank-Nullity theorem for the $T: R^3 \to R^3$ defined by $T(x,y,z) = (x+2y, y-z, x+2z)$.

Inner Product Spaces and Orthogonality.

Inner Product Space :—

Let V be a real vector space. Suppose to each pair of vectors $u, v \in V$ there is assigned a real number, denoted by $<u, v>$. This function is called a (real) inner product on V if it satisfies the following axioms:

(i) Linear Property : $<au_1 + bu_2, v> = a<u_1, v> + b<u_2, v>$

(ii) Symmetric Property: $<u, v> = <v, u>$

(iii) Positive Definite Property : $<u, u> \geq 0$; and $<u, u> = 0$ if and only if $u = 0$.

Norm of a vector :

An inner product, $<u, u>$ is nonnegative for any vector u.

$$\|u\| = \sqrt{<u, u>} \quad \text{or} \quad \|u\|^2 = <u, u>$$

This non negative number is called the norm or length of u.

* Every non zero vector v in V can be multiplied by the reciprocal of its length to obtain the unit vector

$$\hat{v} = \frac{1}{\|v\|} v$$

which is a positive multiple of v.

This process is called normalizing v.

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Orthogonality :-

Let V be an inner product space. The vectors $u, v \in V$ are said to be orthogonal and u is said to be orthogonal to v if

$$\langle u, v \rangle = 0.$$

Problems :—

1) Consider vectors $u = (1,2,4)$, $v = (2,-3,5)$, $w = (4,2,-3)$ in R^3. Find a) $\langle u \cdot v \rangle$ b) $\langle u \cdot w \rangle$ c) $\langle v \cdot w \rangle$ d) $\langle (u+v) \cdot w \rangle$

e) $\|u\|$ f) $\|v\|$

a) $\langle u \cdot v \rangle = 2 - 6 + 20 = 16$

b) $\langle u \cdot w \rangle = 4 + 4 - 12 = -4$

c) $\langle v \cdot w \rangle = 8 - 6 - 15 = -13$.

d) $\langle (u+v) \cdot w \rangle = (3, -1, 9) \cdot (4, 2, -3) = 12 - 2 - 27 = -17$.

e) $\|u\| = \sqrt{1^2 + 2^2 + 4^2} = \sqrt{1 + 4 + 16} = \sqrt{21}$.

f) $\|v\| = \sqrt{4 + 9 + 25} = \sqrt{38}$.

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2) Consider the vectors $u = (1,5)$ and $v = (3,4)$ in R^2;

Find : a) $\langle u, v \rangle$ with respect to the usual inner product in R^2.

b) $\|v\|$ using the inner product in R^2.

3) Consider the following polynomials in $P(t)$ and inner product:

$f(t) = t+2$, $g(t) = 3t-2$, $h(t) = t^2-2t-3$ and

$<f,g> = \int_0^1 f(t)g(t)\,dt$.

(a) find $<f,g>$ and $<f,h>$ (b) find $\|f\|$ and $\|g\|$

(c) normalize f and g.

Sol^n: (a) $<f,g> = \int_0^1 (t+2)(3t-2)\,dt = \int_0^1 (3t^2+4t-4)\,dt$

$$<f,g> = t^3+2t^2-4t\Big|_0^1 = -1.$$

$$<f,h> = \int_0^1 (t+2)(t^2-2t-3)\,dt = \frac{t^4}{4} - \frac{7t^2}{2} - 6t\Big]_0^1 = \frac{-37}{4}$$

(b) $<f,f> = \int_0^1 (t+2)(t+2)\,dt = \frac{19}{3}$; $\|f\| = \frac{\sqrt{19}}{\sqrt{3}} = \frac{\sqrt{57}}{3}$.

$<g,g> = \int_0^1 (3t-2)(3t-2)\,dt = 1$; $\|g\| = \sqrt{1} = 1$

(c) Since $\|f\| = \frac{\sqrt{57}}{3}$ and g is already a unit vector,

$$\hat{f} = \frac{1}{\|f\|} f = \frac{3}{\sqrt{57}}(t+2)$$

$$\hat{g} = \frac{1}{\|g\|} g = 3t-2.$$

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4) Let $M = M_{2,3}$ with inner product. $<A,B> = tr<B^T A>$

and let $A = \begin{bmatrix} 9 & 8 & 7 \\ 6 & 5 & 4 \end{bmatrix}$ $B = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$ $C = \begin{bmatrix} 3 & -5 & 2 \\ 1 & 0 & -4 \end{bmatrix}$

Find (a) $<A,B>$, $<A,C>$, $<B,C>$

(b) $<2A+3B, 4C>$ (c) $\|A\|$ and $\|B\|$

(a) $\langle A, B \rangle = \sum_{i=1}^{m} \sum_{j=1}^{n} a_{ij} b_{ij}$

$\langle A, B \rangle = 9+16+21+24+25+24 = 119$

$\langle A, C \rangle = 27-40+14+6+0-16 = -9$

$\langle B, C \rangle = 3-10+6+4+0-24 = -21$

(b) $2A+3B = \begin{bmatrix} 21 & 22 & 23 \\ 24 & 25 & 26 \end{bmatrix}$ $\quad 4C = \begin{bmatrix} 12 & -20 & 8 \\ 4 & 0 & -16 \end{bmatrix}$

$\langle 2A+3B, 4C \rangle = 252-440+96+0-416 = -324$

(c) $\|A\|^2 = \langle A, A \rangle = \sum_{i=1}^{m} \sum_{j=1}^{n} a_{ij}^2$, the sum of the squares of all the elements of A.

$\|A\|^2 = \langle A, A \rangle = 9^2+8^2+7^2+6^2+5^2+4^2 = 271 \Rightarrow \|A\| = \sqrt{271}$

$\|B\|^2 = \langle B, B \rangle = 1^2+2^2+3^2+4^2+5^2+6^2 = 91 \Rightarrow \|B\| = \sqrt{91}$

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4) Verify the vectors $u=(1,1,1)$, $v=(1,2,-3)$ & $w=(1,-4,3)$ in R^3 are orthogonal or not.

Soln:- $\langle u, v \rangle = 1+2-3 = 0$, $\langle u, w \rangle = 1-4+3 = 0$, $\langle v, w \rangle = 1-8-9 = -16$

Thus u is orthogonal to v and w,

v & w are not orthogonal.